

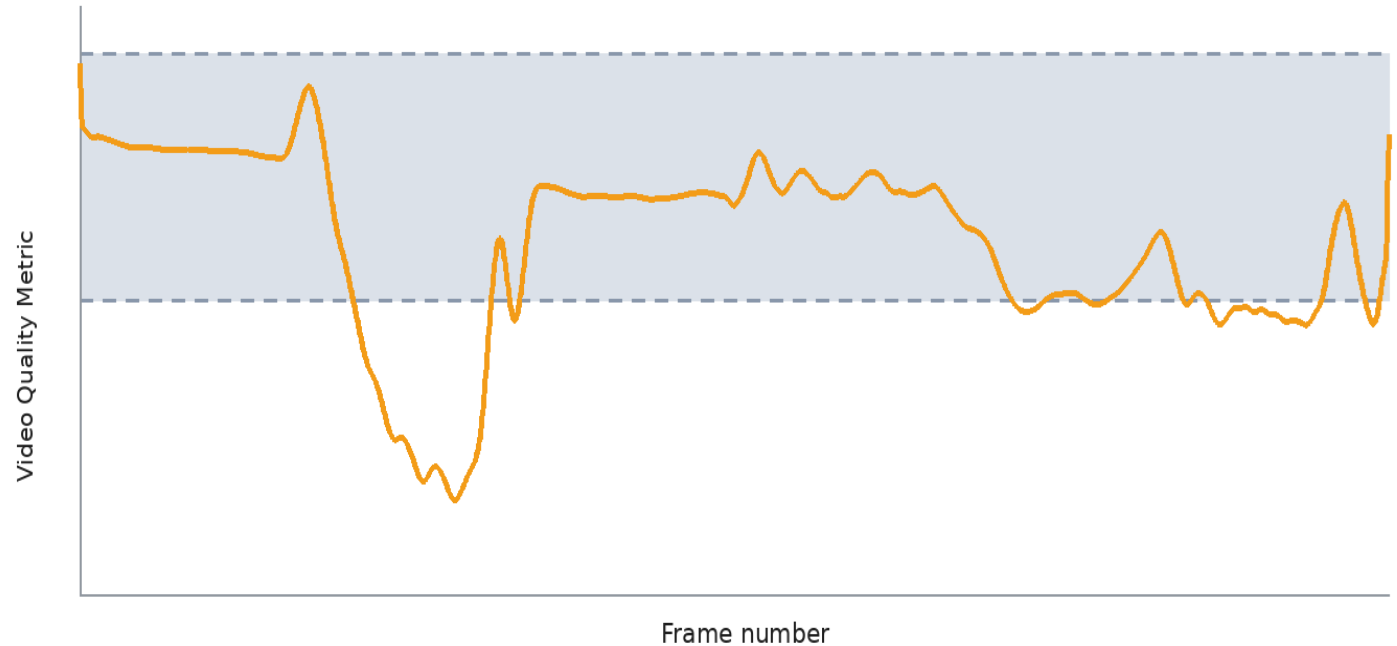
Quality Consistency Score

A survival-based reliability
descriptor for video quality
assessment



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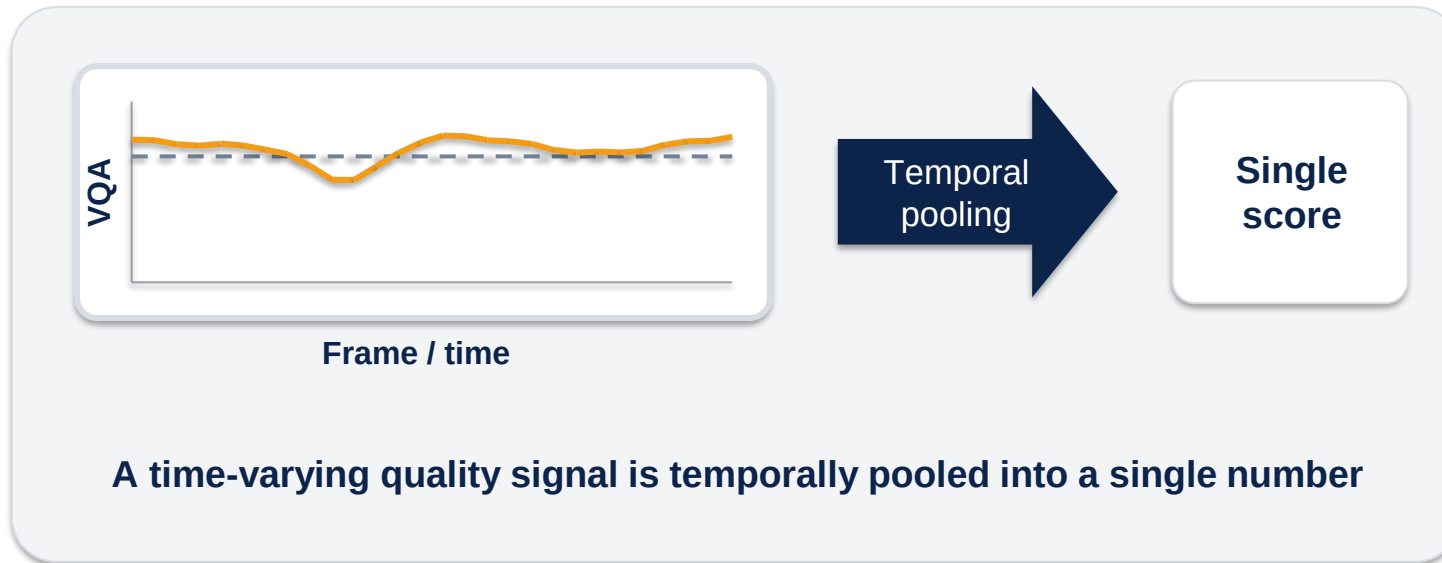
IEEE ICIP 2026
Special session: QoE and Sustainability in
Video Communications



Temporal reliability: quality consistency over time

Motivation: averages can hide quality drops

Real encoder operation on long heterogeneous content is not stationary



Need

A distribution-based view of time-varying quality:
how consistently quality remains above meaningful thresholds over time

Problem: temporal mean pooling collapses when and how often quality drops occur

- Codec benchmarking usually averages per-frame VQA scores for RD/Bjøntegard Difference (BD)-rate
- Effective for short, relatively stationary test sequences (JVET)
- Real content is longer, heterogeneous, and often encoded with rate control
- Rate control may allow temporally quality drops to satisfy global bitrate constraints
- Pooling variants can expose low-quality behavior, but still reduce the temporal distribution to a single summary

Survival curve (S): quality as time-above-threshold

Convert a per-frame quality signal into a distribution over quality thresholds

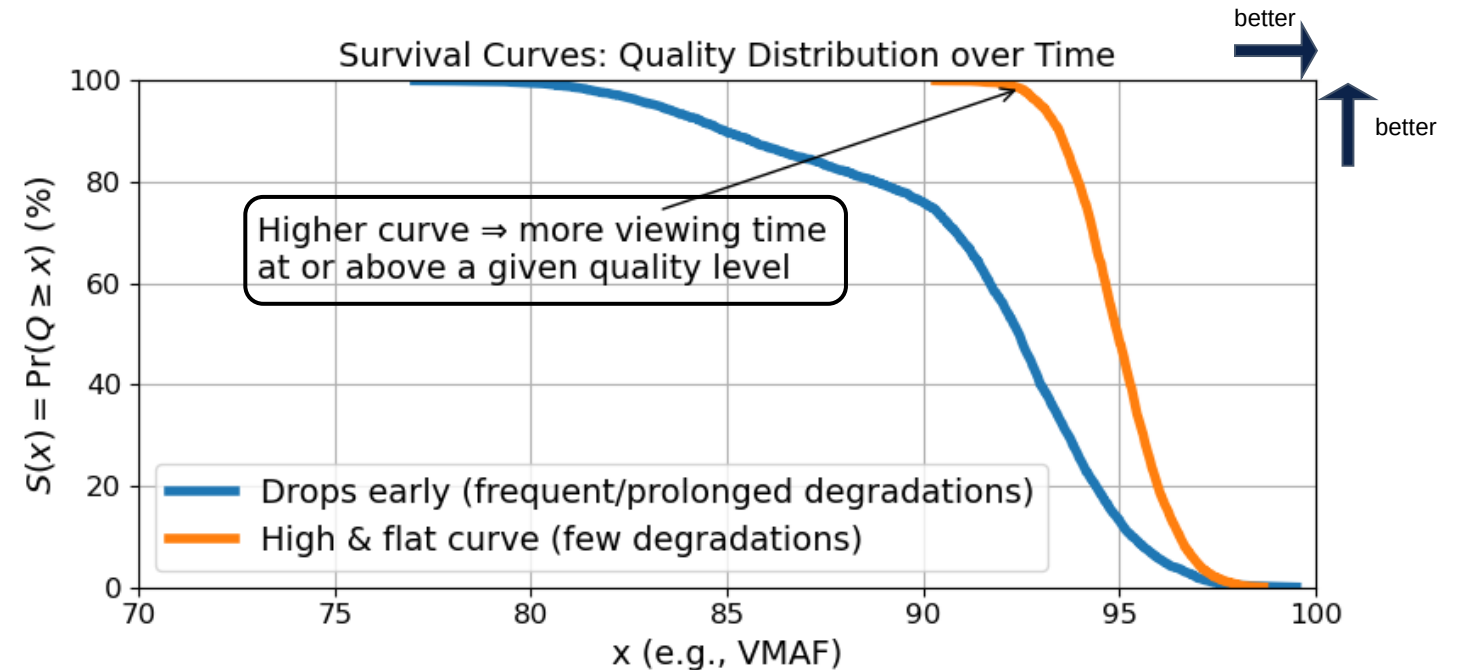
$$S(x) = 100 \cdot \Pr\{Q(t) \geq x\}, \quad S(x) \in [0, 100]$$

$Q(t)$: per-frame quality score

x : VQA threshold

$\Pr\{\bullet\}$: probability over time

$S(x)$: fraction of viewing time with quality $\geq x$



Quality Consistency Score (QCS): range-averaged reliability

Integrate the survival curve over a selected operating range

$$\text{QCS}(S, [a, b]) = \frac{1}{b - a} \int_a^b S(x) dx, \text{QCS} \in [0, 100]$$

Definition

Range-averaged percentage of viewing time at or above thresholds in $[a, b]$

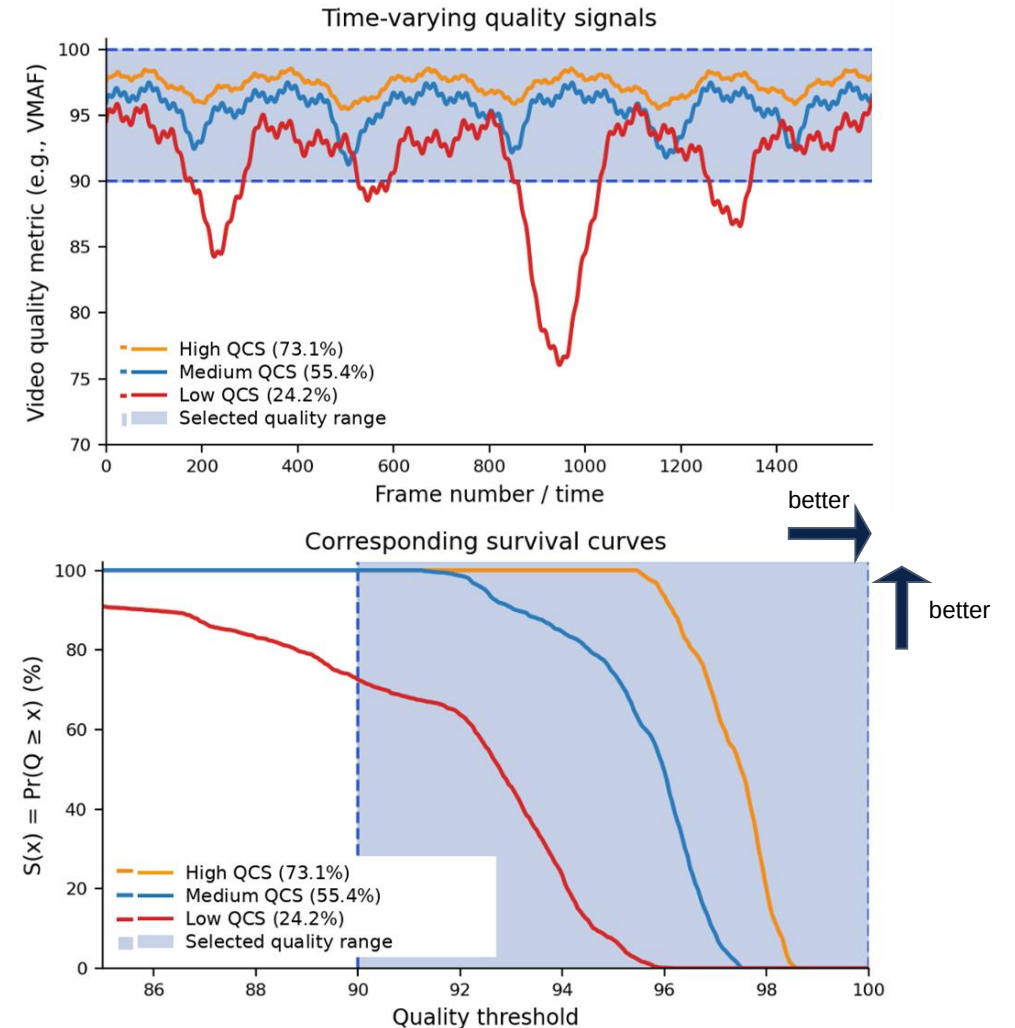
Single threshold

$S(90)$: how often is quality above VMAF 90?

Quality range

$\text{QCS}(S, [90, 100])$: how consistently does quality remain high across VMAF 90 to 100?

Interpretation assuming VMAF



Using QCS correctly

A range-dependent, metric-agnostic descriptor of time-above-threshold reliability

Metric-agnostic post-processing

Apply QCS to any per-frame quality signal: PSNR, MS-SSIM, VMAF, MOS predictors, or subjective frame scores

Range-dependent descriptor

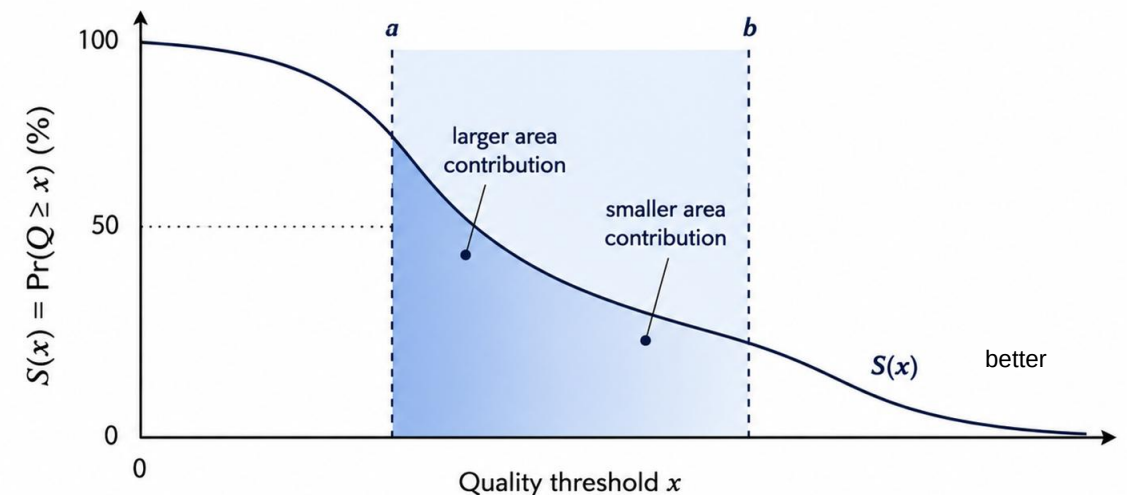
Choose and report [a,b]; QCS values and rankings depend on the selected quality range

Some examples

	PSNR	MS-SSIM	VMAF
Distribution	[30, 50]	[0.8, 1.0]	[70, 100]
Contribution	[40, 50]	[0.9, 1.0]	[90, 100]

Design implication

QCS emphasizes avoiding lower-quality events rather than rewarding gains near transparent quality



$$\int_{c-\varepsilon}^c S(x) dx \geq \int_c^{c+\varepsilon} S(x) dx, \forall c \in [a + \varepsilon, b - \varepsilon]$$

Bjontegard Difference (BD)-QCS: temporal reliability gain at equal bitrate

QCS([a,b]) gives one scalar per bitrate, so BD-style summaries are possible

Explanation

BD-QCS in **percentage points [pp]** is the average QCS difference between test and reference at equal bitrate

Interpretation

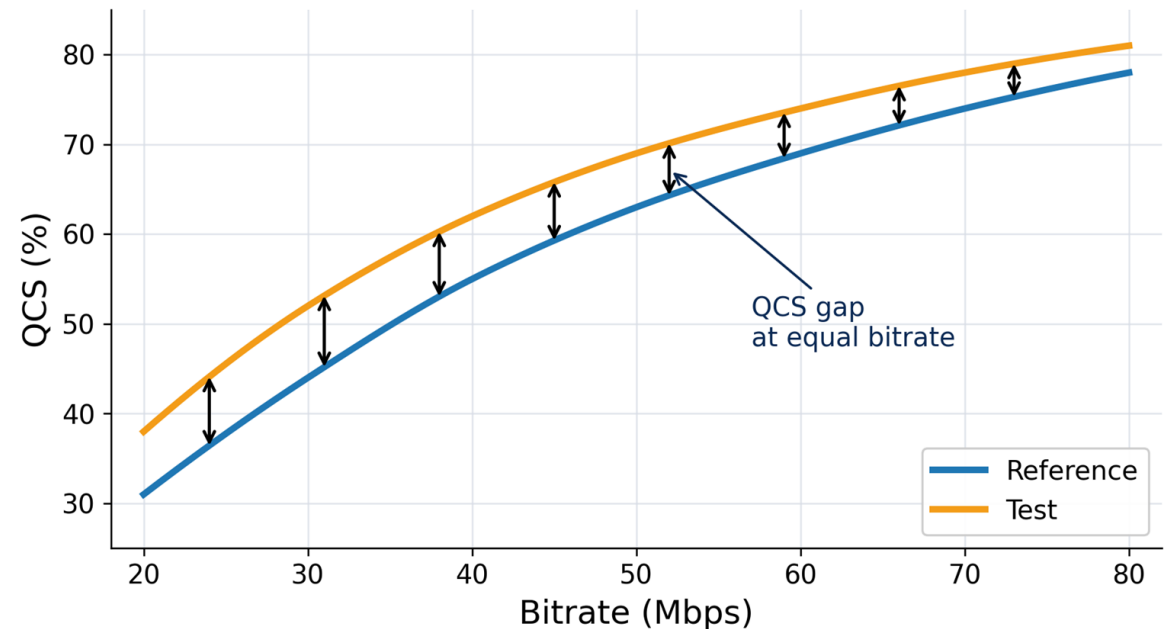
BD-QCS([a,b]) = + Δ pp:
at equal bitrate, Δ additional percentage points of viewing time above the selected quality thresholds, on average

Complements BD-rate

- **BD-rate:** average bitrate savings at equal mean quality
- **BD-QCS:** average temporal-reliability gains at equal bitrate

Average QCS difference at equal bitrate

Vertical black arrows indicate the BD-QCS contribution along the bitrate range



Same bitrate → more viewing time above selected quality thresholds

Where QCS is useful

Several use cases in the video coding workflow

Rate-control comparisons

Compare CBR/VBR, lookahead, adaptive quantization, and quality-preserving modes that may trade local quality for global bitrate constraints.

Adaptive / content-aware encoding

Assess how scene-level adaptivity and per-segment decisions affect the frequency and duration of quality degradations.

Long-form benchmarking

Evaluate heterogeneous content where average pooling can hide intermittent quality drops over time.

QoS-style reporting

Report service-level guarantees as fraction of viewing time above selected thresholds, e.g., VMAF ≥ 90 .

Case study: NVIDIA NVENC codec upgrade

4K UHD HDR live contribution workflow · isolating the codec effect: HEVC reference vs AV1 test



Experimental setup

Rate control

CBR: 20 - 80 Mbps

Live encoding mode

Intra period: 60 frames
Preset p4 · tune hq

VQA metric

Per-frame VMAF

QCS range

VMAF [90, 100]

Six 1-minute 4Kp60 UHD HDR test sequences

- Netflix (x5): BasketballGame, DrivingPOV, Meridian, RollerCoaster, ToddlerFountain
- Fraunhofer HHI (x1): BerlinSeqs

Results: AV1 improves average quality and temporal reliability

BD-QCS makes the reliability gain interpretable as additional high-quality viewing time

	AVG across clips	Interpretation
BD-rate	-11.84%	Average bitrate reduction, at equal mean VMAF
BD-QCS	+8.88 pp	At equal bitrate, additional high-quality viewing time over VMAF [90,100]

+8.88 pp $\approx$ 5.3 s ($=0.088 \times 60$) of additional high-quality viewing time per one-minute clip, on average

Content-dependent gains

Best case

ToddlerFountain:
+39.79 pp $\approx$ 24 s/min

Worst case

RollerCoaster:
+1.11 pp $\approx$ 0.67 s/min

QCS vs. temporal pooling: different questions

BerlinSeqs · AV1 · 35 Mbps · per-frame VMAF summaries

Method	Value	Question answered
Mean pooling	95.68	average quality?
Minkowski $p=0.5$	95.66	Low-quality-weighted score?
p10	92.29	One lower-tail percentile?
Worst-1% mean	88.33	Severity of the worst tail?
QCS[90,100]	57.12%	Range-averaged temporal reliability

Pooling summaries

Mean, Minkowski, percentiles, and worst-k% mean reduce the temporal distribution to one selected score or tail statistic.

QCS

QCS[90,100] = 57.12% is the expected fraction of viewing time above a uniformly sampled threshold in [90,100].

Complement, not replacement: pooling asks “what score?”; QCS asks “how reliably above thresholds?”

Takeaways

QCS complements RD/BD-rate analysis with temporal reliability

1

QCS describes temporal reliability

VQA metric-agnostic descriptor obtained by integrating the survival function of per-frame quality scores over a selected quality range

2

QCS complements RD/BD-rate analysis

Reports the fraction of viewing time spent at or above perceptually meaningful quality levels

3

BD-QCS makes reliability gains interpretable in viewing-time units

Percentage-point gains correspond to additional high-quality viewing time

What QCS does not capture

- Short-term oscillations (temporal flicker)
- Temporal structure of degradations: one long drop vs many brief drops

Future work

- Extend QCS analysis with descriptors of temporal degradation patterns
- Run subjective tests on perceived temporal inconsistencies and correlate results with QCS

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Python code and examples on GitHub



github.com/pixop/Quality-Consistency-Score

Questions?